Textbook Exercises: Modern Classical Physics

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Abstract

Working through the Track 1 (initially, unless a topic is intersting/relevant) problems of [Thorne and Blandford, 2017] as an ~evening project, when motivated, can be bothered, and when it doesn't interfere with my research too much :) .

The bulk of the content and learning is to be done on paper so I won't be laying out condensed notes from the chapters here; although, when neccessary figures may be added, and crosslinks of equations from previous problems if an example builds up the theory.

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3 Kinetic Theory

Exercise 3.3: **Regimes of Particulate and Wave-Like Behavior

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a) *C*ygnus X-1 is a source of X-rays that has been studied extensively by astronomers. The observations (X-ray, optical, and radio) show that it is a distance r ∼ 6,000 light-years from Earth. It consists of a very hot disk of X-ray-emitting gas that surrounds a black hole with mass 15M , and the hole in turn is in a binary orbit with a heavy companion star. Most of the X-ray photons have energies E ∼ 2 keV, their energy flux arriving at Earth is $F \sim 10 \times 10 W\, m^{-2}$, and the portion of the disk that emits most of them has radius roughly 7 times that of the black hole (i.e., $R \sim 300$ km).5 Make a rough estimate of the mean occupation number of the X-rays photon states. Your answer should be in the region $\eta \approx 1$, so the photons behave like classical, distinguishable particles. Will the occupation number change as the photons propagate from the source to Earth?

   Solution

b) *A* highly nonspherical supernova in the Virgo cluster of galaxies (40 million lightyears from Earth) emits a burst of gravitational radiation with frequencies spread over the band $0.5 \sim 2.0 kHz$, as measured at Earth. The burst comes out in a time of about 10 ms, so it lasts only a few cycles, and it carries a total energy of roughly $10^{-3} Mc^2$, where $M = 2x10^{30} kg$ is the mass of the Sun. The emitting region is about the size of the newly forming neutron star core (10 km), which is small compared to the wavelength of the waves; so if one were to try to resolve the source spatially by imaging the gravitational waves with a gravitational lens, one would see only a blur of spatial size one wavelength rather than seeing the neutron star. What is the mean occupation number of the burst's graviton states? Your answer should be in the region $\eta \sim 1$, so the gravitons behave like a classical gravitational wave.

   Solution

Exercise 3.4: **Maxwell Velocity Distribution

Consider a collection of thermalized, classical particles with nonzero rest mass, so they have the Boltzmann distribution:

$$\eta = \exp(-(E - \mu)/k_B T) \tag{3.1}$$

Assuming that the temperature is low enough ($k_B T \ll mc^2$) that they are nonrelativistic.

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1. *Explain why the total number density of particles n in physical space (as measured in the particles' mean rest frame) is given by the integral $n = \int \mathcal{N} dV_p$. Show that $n \propto \exp(\mu/k_B T)$, and derive the proportionality constant.*

   Solution:

The distribution function, $\mathcal{N}$, is defined as the number density of particles in 6D phase space:

$$\mathcal{N} = \frac{dN}{dV_x dV_p} \tag{3.2}$$

Integrating over momentum space gives $\int \mathcal{N} dV_p = \int \frac{dN}{dV_x}$, where the RHS is the number of particles dN in a small volume dV_x, i.e. number density:

$$n = \int \mathcal{N} dV_p \tag{3.3}$$

The distribution function is related to the mean occupation number by:

$$\eta = \frac{\mathcal{N}}{\mathcal{N}_{states}} = \frac{h^3}{g_s}\mathcal{N} \tag{3.4}$$

For a thermalised gas, the distribution function is isotropic in phase space. Using (3.1), we integrate (3.3) over a small volume element in momentum space $dV_p = 4\pi p^2 dp$.

$$\eta = \frac{4\pi h^3 \exp(\mu/k_B T)}{g_s} \int \exp(-E/k_B T) p^2 dp \tag{3.5}$$

with $E = p^2/2m$ the integral, has the following solution

$$\int_0^\infty \exp(-ap^2) p^2 dp = \frac{\sqrt{\pi}}{4} \frac{1}{a^{3/2}} \tag{3.6}$$

where $a = 1/2mk_B T$.

The proportionality constant for $n \propto \exp(\mu/k_B T)$:

$$A = \frac{2\sqrt{2}\pi^{3/2} h^3 (mk_B T)^{3/2}}{g_s} \tag{3.7}$$

2. *Explain why the mean energy per particle is given by* $\bar{E} = n^{-1} \int (p^2/2m) \mathcal{N} dV_p$. *Show that* $\bar{E} = 3/2k_B T$.

Solution:

The integral in the expression for $\bar{E}$ represents the energy density.

Writing all terms out explicitly as integrals, we can cancel out constants to simplify the algebra

$$\bar{E} = \frac{1}{2m} \frac{\int p^4 \exp(-ap^2) dp}{\int p^2 \exp(-ap^2) dp} \tag{3.8}$$

Constant (a) defined as above and $\int p^4 \exp(-ap^2) dp = \frac{3}{8} \frac{\sqrt{\pi}}{a^{5/2}}$ is solved by integration by parts.

Giving an average energy of a particle $\bar{E} = (3/2) k_B T$.

3. *Find $P(v)dv \equiv$ (probability that a randomly chosen particle will have speed $v = \|v\|$ in the range dv).*

   Solution:

   We are considering a non-relativistic gas, so we can work in terms of a particles momentum p and substitute $dp = mdv$ at the end.

   The probability a randomly chosen particles will have a momentum p in the range dp is the number density of particles in the region of momentum space $d\mathcal{V}_p$ defined by $p + dp$, normalised to the total number density:

$$P(p)dp = \frac{\mathcal{N} d\mathcal{V}_p}{\int \mathcal{N} d\mathcal{V}_p} \tag{3.9}$$

   Using the expression for n derived in (3.7) and some algebra gives:

$$P(p)dp = \frac{4\pi p^2 e^{-p^2/2mk_BT} dp}{(2\pi m k_B T)^{3/2}} \tag{3.10}$$

   Recasting the differentials in terms of velocity, and defining a characteristic velocity $v_0^2 = \frac{m}{2k_BT}$:

$$P(v)dv = \frac{4}{\sqrt{\pi}} \frac{v^2}{v_0^3} \exp(-v^2/v_0^2) dv \tag{3.11}$$

4. *Consider particles confined to move in a plane or in one dimension (on a line). What is their speed distribution $P(v)$ and at what speed does it peak?*

   Solution:

   Eq. (3.9) in 1D $d\mathcal{V}_p = dL_p$ gives:

$$P(p)dp = \frac{\exp(-ap^2)dp}{\int \exp(-ap^2)dp} \tag{3.12}$$

   where $a = \frac{1}{2mk_BT}$.
   we can evaluate the above wrt speed v, considering the characteristic velocity above v_0 and the gaussian integral $\int_{-\infty}^{\infty} \exp(-x^2)dx = \sqrt{\pi}$:

$$P(v)dv = \frac{1}{\sqrt{\pi} v_0} \exp(-v^2/v_0^2) dv \tag{3.13}$$

   The velocity distribution is a gaussian centered on zero. When motion is confined to 1D, the most probable speed of a particles is zero.

Exercise 3.8: ! Derivation and Practice: Equation of State for Nonrelativistic, Classical Gas

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Exercise 3.9: Equation of State for Nonrelativistic, Electron-Degenerate Hydrogen

Derive Eq. (3.43) for the electron pressure in a nonrelativistic, electron-degenerate hydrogen gas.

Solution:

The distribution function for electrons (fermions) is given by the Fermi-Dirac distribution:

$$\eta_e = \frac{1}{\exp((E-\mu)/k_B T) + 1} \tag{3.14}$$

For a gas in thermal equilibrium, the distribution function depends on the magnitude of the momentum and is isotropic. The pressure P and number density n are given by:

$$P = \frac{4\pi}{3m} \int_0^\infty \mathcal{N} p^4 dp \tag{3.15}$$

$$n = 4\pi \int_0^\infty \mathcal{N} p^2 dp \tag{3.16}$$

where $\mathcal{N} = (2/h^3)\eta_e$ and $g_s = 2$.

Given the gas is degenerate ($\mu_e \gg k_B T$), at a sufficiently low temperature, almost all states are fully occupied ($\eta_e \approx 1$), up to an energy state specified by the Fermi-momentum $p_f = \sqrt{2m_e\mu_e}$. Integrating up to the Fermi-momentum, and noting that the electron pressure dominates over the proton pressure $P \approx P_e$, given the mass hierarchy $m_p \gg m_e$:

$$P_e = \frac{8\pi}{15h^3} \frac{p_f^5}{m_e} \tag{3.17}$$

$$n_e = \frac{8\pi}{15h^3} p_f^3 \tag{3.18}$$

For a Hydrogen gas $n_e = n_p = n$, and $m_p \gg m_e$, the density is given by $\rho = m_p n = m_p n_e$. Using this to rearrange p_f in terms of the density ρ and substituting into Eq. (3.17), we get the equation of state:

$$P = \frac{1}{20} \left(\frac{3}{\pi}\right)^{2/3} \frac{m_e c^2}{\lambda_c^3} \left(\frac{\rho}{m_p/\lambda_c^3}\right)^{5/3} \tag{3.19}$$

where $\lambda_c = h/m_e c^2 \approx 10^{-10}$ cm is the Compton wavelength.

Exercise 3.11: Specific Heat for Nonrelativistic, Degenerate Electrons in White Dwarfs and in Metals

Consider a nonrelativistically degenerate electron gas at finite but small temperature.

a) *Show that the inequalities $k_BT \ll \mu_e \ll m_e$ are equivalent to the words "nonrelativistically degenerate."*

Solution:

Non-relativistic: The typical kinetic energy of a particle $\sim k_BT$ is much less than the rest-mass energy m_e. Equivalently, the typical velocities of the gas are less than the speed of light $v \ll c$. Degenerate: Given the condition $k_BT \ll \mu_e$, and the fermi-dirac distribution (3.14), energy states $E < \mu_e$ ($E > \mu_e$) are either fully occupied (empty) $\eta_e \approx 1$ ($\eta_e \approx 0$)

b) *Show that the electron mean occupation number $\eta_e(E)$ has the form depicted in Fig. 1: It is near unity out to (nonrelativistic) energy $E \sim \mu_e - 2k_BT$, and it then drops to nearly zero over a range of energies $\Delta E \sim 4k_BT$.*

Solution:

$\eta_e = 1/(\exp(\pm 2) + 1) \approx 0.12(0.88)$

c) *If the electrons were nonrelativistic but nondegenerate, their thermal energy density would be $U = (3/2)nk_BT$, so the total electron energy (excluding rest mass) in a volume V containing $N = nV$ electrons would be $E_{tot} = (3/2)Nk_BT$, and the electron specific heat, at fixed volume, would be*

$$C_v \equiv (\frac{\partial E_{tot}}{\partial T})_V = \frac{3}{2}Nk_B \text{ (nondegenerate, nonrelativistic)} \tag{3.20}$$

Using the semiquantitative form of η_e depicted in Fig. 3.8, show that to within a factor of order unity the specific heat of degenerate electrons is smaller than in the nondegenerate case by a factor $\sim k_BT/\mu_e$:

$$C_v \equiv (\frac{\partial E_{tot}}{\partial T})_V \sim \frac{k_BT}{\mu_e}Nk_B \text{ (degenerate, nonrelativistic)} \tag{3.21}$$

Solution:

As seen on Fig. 1, only electrons around $E \approx \mu_e$ over a range $\Delta E \approx 4k_BT$, when absorbing energy $\sim k_BT$ can be excited into unoccupied states and therefore contribute to the specific heat. The number, relative to the non-degenerate case can be estimated by taking the ratio of the energy widths

$$N_{avail} = \frac{\Delta E}{\mu_e}N \sim \frac{k_BT}{\mu_e}N \tag{3.22}$$

Given that each electron contributes roughly k_BT of energy, the total energy of the gas $E_{tot} \sim (\frac{k_BT}{\mu_e})Nk_BT$, taking the partial derivative at a fixed volume gives eq. (3.21), where $C_v \propto T$.

d) *Compute the multiplicative factor in Eq. (3.60) for C_V . More specifically, show that, to first order in k_BT/μ_e,*

$$C_V = \pi^2/2(k_BT/\mu_e)Nk_B \tag{3.23}$$

Solution:

$$U = \frac{g_s}{2mh^3}\int_0^\infty \frac{p^4}{\exp((E-\mu)/k_BT)+1}dp \tag{3.24}$$

With the substitution $p = \sqrt{E/2m_e}$

$$U = \frac{8\sqrt{2}\pi m_e^{3/2}}{h^3} \int_0^\infty E^{3/2}\eta(E)dE \tag{3.25}$$

From here on we will drop the coefficients and working with the integral $\int(E)$.

Integrating by parts with $(u' = E^{3/2}, v = \eta(E))$ the first term vanishes over the limits leaving:

$$-\frac{2}{5}\int_0^\infty E^{5/2}\frac{\partial\eta}{\partial E}dE \tag{3.26}$$

As can be seen on Fig. 1, at low T, the occupation number is constant except for a small region around $(E = \mu)$, making η' a well-localised even function. Expanding $\Phi(E) = E^{5/2}$ about $(E = \mu)$ giving:

$$\int_0^\infty [\Phi(\mu_e) + \Phi'(\mu_e)(E - \mu_e) + \frac{\Phi''(\mu_e)}{2}(E - \mu_e)^2 + ...]\frac{\partial\eta}{\partial E}dE \tag{3.27}$$

Making the subsitition $x = (E - \mu_e)/k_BT$, second term vanishes due to integrating odd function over symmetric limits.

$$= k_BT \tag{3.28}$$

Exercise 3.12: Specific Heat for Phonons in an Isotropic Solid

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In Sec. 12.2 we will study classical sound waves propagating through an isotropic, elastic solid. As we shall see, there are two types of sound waves: longitudinal with frequency-independent speed CL, and transverse with a somewhat smaller frequency-independent speed C_T . For each type of wave, s = L or T , the material of the solid undergoes an elastic displacement $\xi = Af_s \exp(ik\dot{x} - \omega t)$, where A is the wave amplitude, f_s is a unit vector (polarization vector) pointing in the direction of the displacement, k is the wave vector, and ω is the wave frequency. The wave speed is $C_s = \omega/\|k\|$ (= C_L or C_T). Associated with these waves are quanta called phonons. As for any wave, each phonon has a momentum related to its wave vector by $p = (h/2\pi)k$, and an energy related to its frequency by $E = (h/2\pi)\omega$. Combining these relations we learn that the relationship between a phonon's energy and the magnitude $p = \|p\|$ of its momentum is $E = C_s p$. This is the same relationship as for photons, but with the speed of light replaced by the speed of sound! For longitudinal waves fL is in the propagation direction k, so there is just one polarization, $g_L = 1$. For transverse waves f_T is orthogonal to k, so there are two orthogonal polarizations (e.g., $f_T = e_x$ and $f_T = e_y$ when k points in the e_z direction), $g_T = 2$.

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1. *c) Ignoring the fact that the sound waves' wavelengths $\lambda = 2\pi/|k|$ cannot be smaller than about twice the spacing between the atoms of the solid, show that the total phonon energy (wave energy) in a volume V of the solid is identical to that for blackbody photons in a volume V , but with the speed of light c replaced by the speed of sound Cs, and with the photon number of spin states, 2, replaced by gs = 3 (2 for transverse waves plus 1 for longitudinal): Etot = asT 4V , with $as = g_s(4\pi 5/15)(k_B^4/(h^3C_s^3))$ [cf. Eqs. (3.54)]*

   Solution:

Distribution function in blackbody (planck form)

$$\mathcal{N} = \frac{g_s}{h^3} \frac{1}{\exp(E/k_B T) - 1} \tag{3.29}$$

$$E_{tot} = V \int n E_p dV_p \tag{3.30}$$

$$= 4\pi g_s C_s V \int_0^\infty \frac{p^3}{\exp(C_s p/k_B T) - 1} dp \tag{3.31}$$

Solved with integral of form ($\int_0^\infty \frac{x^3}{\exp(ax)-1} dx = \frac{7\pi^4}{120a^4}$), with $a = C_s/k_B T$.

References

Kip S Thorne and Roger D Blandford. *Modern Classical Physics: Optics, Fluids, Plasmas, Elasticity, Relativity, and Statistical Physics*. Princeton University Press, 2017.

A Figures

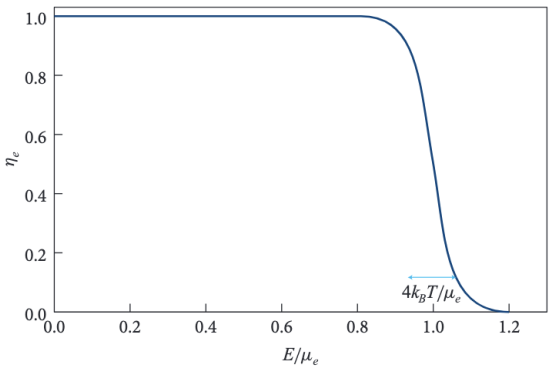

FIGURE 3.8 The Fermi-Dirac distribution function for electrons in the nonrelativistic, degenerate regime $k_BT \ll \mu_e \ll m_e$, with temperature such that $k_BT/\mu_e = 0.03$. Note that η_e drops from near 1 to near 0 over the range $\mu_e - 2k_BT \lesssim E \lesssim \mu_e + 2k_BT$. See Ex. 3.11b.

Figure 1: *